TESTED AND HOPED ON

control software for the physical world, and the language that proves it

Nobody can prove a controller will never reach the network while it is steering a seven-ton flywheel. It is tested, reviewed, and hoped on.

Agents now write that software. The grid is where it bites first.

WHY NOW

Coding agents write production control code this year.

WHY THE GRID

Every new inverter is a software product under UL 1741 SB, and Utah's data centers are the load.

WHY HERE

Torus is scaling a factory in Salt Lake; Hill AFB verifies flight software by review.

MENTL

a language whose compiler proves what a program can never do

NEVER

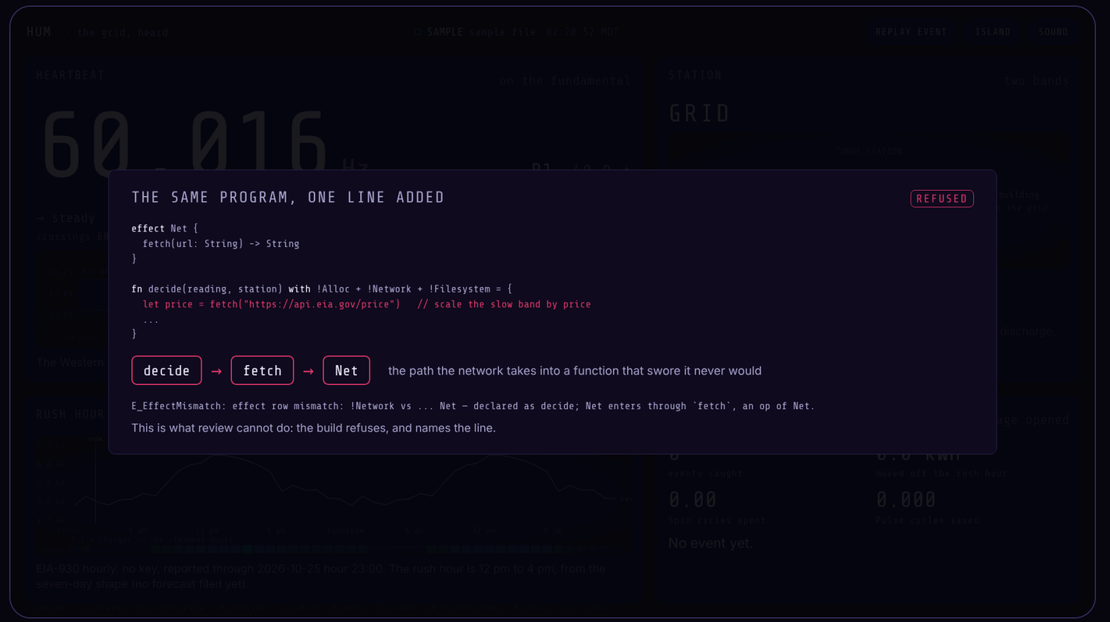
"This function never reaches the network, never allocates, never exceeds this bound" is a claim the compiler proves through every call, or the program does not build.

WHY

Every fact carries its reason, walkable to the line that decided it. A dispatch explains itself.

LEARNS

A learning rule is read off the chain it learns on, no tape, no second copy to drift.



Self-hosting: the pinned compiler reproduces itself byte for byte. One WebAssembly file. Open source. [[years]] of commits.

BUILT THIS WEEK

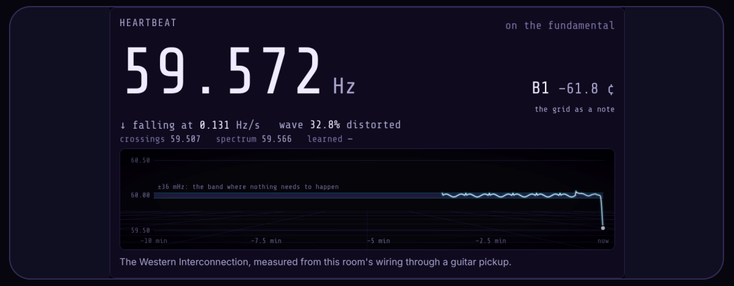
beside what existed before

BEFORE KICKOFF

- Mentl: the compiler, the proofs, the course, the Wasatch piece made of Salt Lake's readings
- sonido: 36 effects, the plugin, the browser editor, the pedal firmware

SINCE MONDAY 4 PM

- The ear: grid frequency from a guitar, three ways, to the millihertz
- The controller in Mentl, its proofs, its refusal twin, the learner
- The page, the director, the data, the real event replay
- The pedal as a second ear; an agent's proposal refused by the gate



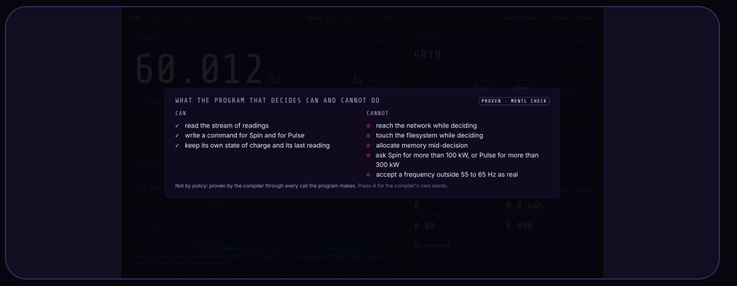
The Western Interconnection, measured from this room's wiring through a guitar pickup.



"falling at 131 mHz per second, past 100"

Modeled from Torus's published figures: 10C Spin, six-minute discharge, four-hour Pulse, under 250 ms. Simulated.

SPIN · PULSE · LASSO



./LEDGER.SH PRINTS EVERY COMMIT WITH ITS TIME

WHO NEEDS IT, IN UTAH

TORUS

Their Station hears every cycle. Their security is a perimeter. The program itself has never been provable, until now. The Torus judge heard this demo today.

HILL AFB, 309TH SOFTWARE ENGINEERING GROUP

Utah's largest employer of software developers, verifying flight software by review. A proof is a deliverable they can file.

EVERY UTAH COMPANY PUTTING AGENTS ON PRODUCTION CODE

A gate that answers PROVEN or REFUSED, with the line, before anything an agent wrote runs.

First hires: compiler and controls engineers, Salt Lake City.

WHAT $10,000 UNLOCKS

ninety days, two introductions, one co-founder

Day 30

The proven controller on inverter-class hardware on my bench. SAM.gov and SBIR.gov registered. Company formed. [[date]]

Day 60

One Utah site, one ear on a real Station-class inverter, the dispatch explained to its operator. The SBIR Phase I draft with Nucleus Grow.

Day 90

Ninety days of decisions on the page, the proposal filed, the first paid pilot signed.

The business: paid pilots with asset operators, a hosted proof gate for teams running coding agents, the SBIR lane for software assurance.

THE $10,000

$4,000 inverter-class hardware and a GPS clock
$3,000 ninety days of the proof gate hosted
$3,000 SBIR preparation and travel

What I need from you:
an introduction at Torus engineering
an introduction at the 309th
a co-founder with power-electronics depth

THE TRUSTED BASE

- The compiler is written in Mentl and the pinned binary reproduces itself byte for byte; that fixed point is the release.
- The verifier is sound for what it decides and incomplete by design: a claim it cannot decide is reported as pending, never silently passed.
- One false proof is known and documented: a recursion deep enough to exhaust the stack traps although its function says it cannot.
- Rust prevents memory bugs, not effects; Simulink and SCADE verify models and generate C. Neither states "never reaches the network" as an obligation through every callee, proves a function allocation-free, or derives a learning rule from the chain.

RUNNING ON A CONTROLLER

- The proof is at compile time. The artifact is one WebAssembly module importing WASI and nothing else.
- It runs wherever wasmtime runs, which includes the Linux supervisors that sit beside inverters today; the demo's controller runs that way.
- A native backend is on the roadmap; the pedal in the demo runs the ear natively on a Cortex-M7 today.
- The learner and the predictor keep learning inside a loop proven not to allocate: learning at the edge that cannot stall.

DURABLE COMPUTATION

A running controller is a value: persisted by copying its image mid-stream and resumed with mentl resume, continuing from the same tick with the same state of charge. The recording is in the repository; the limits are stated there too.

When the grid goes dark, the computation does not.

THE MESH

MESH

SIMULATED



Two Stations with GPS time see the **angle** between them, which is the flow on the line between them, and an event's wave arrives at one before the other. FNET has done this from wall outlets since 2004; a fleet that both sees and answers at 10C has not existed. Simulated from the day's shape and the replayed event.